Reaction-Diffusion Systems and Pattern Formation

5592286 - MA262

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1 Introduction

Nature loves symmetry.
From birth, embryos exist as approximately symmetric clumps of simple cells, dividing uniformly, into larger and larger symmetric cell clumps.

And yet, at the same time, natural processes seem to frequently evolve into structured, non-uniform patterns. For example, animals exhibit a vast variety of patterns on their skin, from zebra stripes and leopard spots, to the giant pufferfish's maze-like patterns.



Figure 1: Giant pufferfish skin pattern[1]

This observation raises a natural question: how can systems starting from approximately uniform states, governed by symmetric laws evolve into asymmetric and highly structured patterns?

These patterns, known as "Turing Patterns", were first introduced by Alan Turing in *The Chemical Basis of Morphogenesis* [7]. In this essay, we explore the mathematical basis of the "reaction-diffusion" systems, the idea of Turing Instability — which bridges the gap between our intuition and reality — and the various conditions that give rise to many familiar patterns within these systems.

2 The General Reaction-Diffusion System

The underlying mechanisms we want to study are reaction-diffusion systems. In these systems, some chemicals, or "reactants", chemically react into "products". These products then diffuse across our medium and may behave as reactants in other reactions. Notably, this scenario could serve as a reasonable guess for the conditions on animal prints.

In order to derive a mathematical model for these systems, it is necessary to approach them from two directions — we must derive both reaction and diffusion mechanics separately, and combine the ideas into one model afterwards.

[1]Taken from Chiswick Chap, CC BY-SA 3.0 ¡https://creativecommons.org/licenses/by-sa/3.0¿, via Wikimedia Commons

2.1 Derivation of Reaction Equations

The reaction side is fairly simple, we know enough from MA146 to formulate the following definition.

Definition 2.1 (Law of Mass Action) *Consider the following reaction:*

$$\mathrm{m_1U_1 + m_2U_2 + \ldots + m_MU_M \xrightarrow{k} n_1U_1 + n_2U_2 + \ldots + n_MU_M},$$

with $m_i,\ n_i \in \mathbb{N} \cup \{0\},\ i = 1, 2, \ldots, M$, *and* $U_1, U_2, \ldots, U_M$ *some substances. The rate of change of concentration of any substance* $U_i,\ i = 1, 2, \ldots, M$ *with concentration of* U_i *denoted by* $u_i(t) : [0, \infty) \to \mathbb{R}$, *is given by:*

$$\frac{du_i}{dt} = R_{U_i}(U_1, U_2, \ldots, U_M) = (n_i - m_i)k \prod_{j=1}^{M} u_j(t)^{m_j},$$

where we call R_{U_i} *the rate function for* U_i *in this reaction.*

Importantly, the rate function for any substance is proportional to the concentrations of all reactants involved — which implies the rate function should be considered to be **non-linear**.

Thus, for a case with n substances, we write a vector function $\mathbf{u} = [0, \infty) \to \mathbb{R}^n$ containing all concentrations:

$$\mathbf{u}(t) := \begin{pmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_n(t) \end{pmatrix}.$$

We can then write the vector rate function of $\mathbf{u}$, $\mathbf{R} : \mathbb{R}^n \to \mathbb{R}^n$, as:

$$\mathbf{R}(\mathbf{u}) = \begin{pmatrix} R_{y_1}(y_1, y_2, \cdots, y_n) \\ R_{y_2}(y_1, y_2, \cdots, y_n) \\ \vdots \\ R_{y_n}(y_1, y_2, \cdots, y_n) \end{pmatrix}. \tag{1}$$

2.2 Derivation of Diffusion Equations

This section is dedicated to proving the following theorem:

Theorem 2.2 (Diffusion Equation) *Let* $\mathbf{u}(\mathbf{x}, t) \in \mathbb{R}^n$ *be a vector describing the concentration of* n *substances at time* $t \in [0, \infty)$ *and position* $\mathbf{x} \in \Omega$, *where* $\Omega \subset \mathbb{R}^3$. *Then, let* D *be a diagonal diffusion matrix, and we obtain the diffusion equation:*

$$\frac{\partial \mathbf{u}}{\partial t} = D\nabla^2\mathbf{u}. \tag{2}$$

In order to do this, we must assume that our diffusion process obeys Fick's Diffusion Laws as outlined in [3, pp. 31-32].

Definition 2.3 (Fick's First Law) *Let y be the concentration of solute in a solution at a height $x \in \mathbb{R}$ from the bottom of a container. Assume y is a decreasing function of x — such that as height increases, concentration of solute decreases or remains constant.*
Now, let $J(x)$ be the diffusive flux at position x. Then, assuming the volume containing the solution does not change, we have:

$$J = -k\frac{dy}{dx},$$

with k the diffusive constant dependent on physical characteristics of the substances.

This law essentially acts as a assumption — that assumption being that the rate of movement of the molecules of the solute at a particular position (flux) is proportional to the change in concentration of the solute at that point.

Furthermore, the constant of proportionality is negative in order to ensure that molecules move away from areas of high concentration, and into adjacent areas of lower concentration, such that the net movement of molecules is from high concentration to low - exactly as we'd expect physically.



Figure 2: An illustration of Fickian diffusion - the purple solute is in high concentration on the left, with the lower concentration to its right. Over time, the solute diffuses to fill the container[1]

Now, extending this idea to 3 spatial dimensions, we get that the flux is just proportional to the gradient of concentration of solute, which is given by:

$$J = -k\nabla y. \tag{3}$$

We are now ready to prove Theorem 2.2

Proof of Theorem 2.2 We begin by considering a model for the movement of particles. Luckily, we already have an equation that works for our case, that being the generalised continuity (transport) equation (from MA265 — generalised to multiple spatial dimensions):

$$\frac{\partial y}{\partial t} + \nabla \cdot J = 0.$$

[1]Adapted from Sbyrnes321, Public domain, via Wikimedia Commons

We can then simply use (3) to give us;

$$\frac{\partial y}{\partial t} + \nabla \cdot (-k \nabla y) = 0,$$

then, since divergence is a linear operator:

$$\frac{\partial y}{\partial t} - k(\nabla \cdot \nabla y) = 0,$$

but, of course, $\nabla \cdot \nabla y = \nabla^2 y$ where ∇^2 is the Laplacian operator. So:

$$\frac{\partial y}{\partial t} = k \nabla^2 y.$$

However, we want a solution involving $\mathbf{n}$ such solutes. Denoting the concentrations of each of these by $y_i,\ i = 1, 2, \ldots, n$, with respective diffusive coefficients $k_i,\ i = 1, 2, \ldots, n$ we may write

$$\mathbf{u} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix},$$

and particularly, we may write our diffusion equation above for all our solutes together as:

$$\frac{\partial \mathbf{u}}{\partial t} = \begin{pmatrix} k_1 \nabla^2 y_1 \\ k_2 \nabla^2 y_2 \\ \vdots \\ k_n \nabla^2 y_n \end{pmatrix} = \begin{pmatrix} k_1 & 0 & 0 & \ldots \\ 0 & k_2 & 0 & \ldots \\ \vdots & & \ddots & \\ 0 & 0 & \ldots & k_n \end{pmatrix} \nabla^2 \mathbf{u}.$$

Then, denoting the diagonal matrix of diffusive coefficients (the "diffusion matrix") as D, we get, finally

$$\frac{\partial \mathbf{u}}{\partial t} = D \nabla^2 \mathbf{u}.$$

□

2.3 The General Reaction Diffusion System

We can now finally write down the general reaction diffusion system, by combining (1) and Theorem (2.2).

Definition 2.4 (Reaction Diffusion System) *Let the vector* $\mathbf{u}(\mathbf{x}, t) : \Omega \to \mathbb{R}^n$*, where* $\Omega \subset \mathbb{R}^3$*, denote the concentration of* n *substances at position* $\mathbf{x}$ *and time* t*. Then,*

$$\frac{\partial \mathbf{u}}{\partial t} = D \nabla^2 \mathbf{u} + \mathbf{R}(\mathbf{u}),$$

where $\mathbf{R}(\mathbf{u})$ *is the vector describing the rate functions of all our substances, and* D *is a diagonal matrix of diffusive coefficients*

For the rest of this essay, we'll consider only **2 component systems**, so systems involving 2 chemicals and reactions between them.

Example (Gray Scott Model) A classic example of a 2 component reaction-diffusion system is the Gray Scott Model (adapted from [4, p. 1087]).
We first assume we're working in a space with constant volume, and that substances within this volume are free to flow in and out of it.
In this space, we consider the following reaction:

$$\begin{aligned} \text{source} &\xrightarrow{\text{F}} \text{U} \\ \text{U} + 2\,\text{V} &\xrightarrow{1} 3\,\text{V} \ , \\ \text{V} &\xrightarrow{\text{k}} \text{sink} \end{aligned}$$

Where U is drawn from a source of fixed concentration 1 (i.e. a fully saturated reservoir of U) — so U is physically brought into the system *changing volume*.
V is chemically disposed of in a sink — so something happens that leads to V becoming something else that we're no longer interested in *but this doesn't change volume*.
Then, with $u(\mathbf{x}, t),\ v(\mathbf{x}, t)$ the concentrations of $U,\ V$ respectively, by considering the Law of Mass Action, and conservation of volume, we can write the rate functions;

$$\begin{aligned} R_U(u, v) &= -uv^2 + F(1 - u), \\ R_V(u, v) &= uv^2 - (F + k)v. \end{aligned}$$

Which, with $\mathbf{u}(\mathbf{x}, t) = \begin{pmatrix} u(\mathbf{x}, t) \\ v(\mathbf{x}, t \end{pmatrix}$ and $D_U,\ D_V$ diffusive constants for $U,\ V$, we get the Gray Scott model:

$$\frac{\partial \mathbf{u}}{\partial t} = \begin{pmatrix} D_U & 0 \\ 0 & D_V \end{pmatrix} \nabla^2 \mathbf{u} + \begin{pmatrix} -uv^2 + F(1 - u) \\ uv^2 - (F + k)v \end{pmatrix}.$$

3 Turing Instability

Now we need to think about the mechanism that leads to our pattern formation. Recall the introduction — we would be starting from a state where chemicals are approximately uniformly distributed across our surface — this is analogous to having a well-mixed mixture of our substances.
This state lacks any sort of regularity, so, patterns must emerge from some small deviations from a homogeneous mixture.

This motivates the following — if we can show that small deviations, or perturbations, from a completely homogeneous, stable state grow into more regular structures in the presence of diffusion — then we have suitably explained the origin of these patterns.

3.1 Homogeneous Steady States

We now want to formalise these ideas. First, defining Turing instability (adapted from [6, p. 76]);

Definition 3.1 (Turing Instability) *Let* $\mathbf{u}(x,t) = (u^*,\ v^*)^T$ *be a homogeneous equilibrium of a 2-component reaction-diffusion system.*
We say the equilibrium exhibits ***Turing instability*** *if it is stable in the kinetic system (*$\mathbf{D} = \mathbf{0}$*), but becomes unstable to non-uniform spatial perturbations in the presence of diffusion.*

This "diffusion-driven" instability forms the basis of the rest of our study; so it's imperative that each component of this definition is well understood.

Definition 3.2 (Homogenous Equilibrium/Steady State) *We say a 2 component reaction-diffusion state is in homogeneous equilibrium if it is in a homogeneous steady state — that is,* $\mathbf{u}(x,t) = (u^*(\mathbf{x},t),\ v^*(\mathbf{x},t))^T$ *and:*

$$0 = \mathbf{D}\nabla^2 \begin{pmatrix} u^* \\ v^* \end{pmatrix} + \mathbf{R}\left(\begin{pmatrix} u^* \\ v^* \end{pmatrix}\right),$$

and more specifically:

1. u^* *and* v^* *do not depend on space (*$u^*(\mathbf{x},t) = u_*,\ v^*(\mathbf{x},t) = v_*\ \forall \mathbf{x}$ *with* $u_*,\ v_* \in \mathbb{R}$*).*
2. *There are no changes to concentrations (*$\mathbf{R}\left((u^*, v^*)^T\right) = 0$*).*

In a chemical context, this is akin to having a particular mixture of substances in which each point in space has the exact same concentration of each chemical, and none of the concentrations change.
Thus, we're in a state of perfect equilibrium, reactions cause no net changes, and our system is **stable**.
This means that in the absence of diffusion, if we were to introduce any small change to the concentrations at any point, the chemical kinetics of the system would bring that point back into that balanced state.

We can't necessarily say the same when diffusion is introduced though. Since each point affects its neighbours, there's no certainty that the system returns to this steady homogeneous state. So, we want to explore what happens near a homogeneous steady state, and see how diffusion interacts with small perturbations.

3.2 Linear Stability Analysis

Here, we follow the analysis as outlined in [5, pp. 3-5].

First, we formulate the general 2-component reaction diffusion system (2.4) as:

$$\frac{\partial \mathbf{u}}{\partial t} = D\nabla^2\mathbf{u} + R(\mathbf{u}) \implies \begin{pmatrix}\partial_t u\\ \partial_t v\end{pmatrix} = \begin{pmatrix}D_U & 0\\ 0 & D_V\end{pmatrix}\begin{pmatrix}\nabla^2 u\\ \nabla^2 v\end{pmatrix} + \begin{pmatrix}R_U(u,v)\\ R_V(u,v)\end{pmatrix}. \tag{4}$$

(Note here we use ∂_t to denote the partial derivative with respect to t)
We'll start with $\mathbf{u}(\mathbf{x},t) = (u^*(\mathbf{x},t),\ v^*(\mathbf{x},t))^T$, a homogeneous steady state by Definition 3.2.
We then consider an initial state close to the homogeneous steady state. So, with $\mathbf{u} = (u,v)^T$ and $\mathbf{u}^* = (u^*,v^*)^T$, we introduce a small, non-uniform perturbation, $\mathbf{U}(\mathbf{x},t)$, with $\mathbf{u} = \mathbf{u}^* + \varepsilon\mathbf{U}$ and $|\varepsilon| \ll 1$. Then using the fact that $\dfrac{\partial}{\partial t}$ and ∇^2 are linear operators:

$$\begin{aligned}\frac{\partial(\mathbf{u}^* + \varepsilon\mathbf{U})}{\partial t} &= D\nabla^2(\mathbf{u}^* + \varepsilon\mathbf{U}) + \mathbf{R}(\mathbf{u}^* + \varepsilon\mathbf{U}),\\ \frac{\partial \mathbf{u}^*}{\partial t} + \varepsilon\frac{\partial \mathbf{U}}{\partial t} &= D(\nabla^2\mathbf{u}^* + \varepsilon\nabla^2\mathbf{U}) + \mathbf{R}(\mathbf{u}^* + \varepsilon\mathbf{U}).\end{aligned}$$

Then since $\mathbf{u}^*$ satisfies the properties given in Definition 3.2:

$$\varepsilon\frac{\partial \mathbf{U}}{\partial t} = D\varepsilon\nabla^2\mathbf{U} + \mathbf{R}(\mathbf{u}^* + \varepsilon\mathbf{U}).$$

We now need to expand the rate function term. Using a Taylor series expansion around $\mathbf{u}^*$:

$$\mathbf{R}(\mathbf{u}^* + \varepsilon\mathbf{U}) = \mathbf{R}(\mathbf{u}^*) + \mathbf{J}(\mathbf{u}^* + \varepsilon\mathbf{U} - \mathbf{u}^*) + O(\varepsilon^2),$$

where $\mathbf{J}$ is the Jacobian matrix of $\mathbf{R}$ at $\mathbf{u}^*$. The $O(\varepsilon^2)$ represents terms of order ε^2 and higher, which we discard as ε is very small, leaving:

$$\varepsilon\frac{\partial \mathbf{U}}{\partial t} = D\varepsilon\nabla^2\mathbf{U} + \mathbf{R}(\mathbf{u}^*) + \varepsilon\mathbf{J}\mathbf{U},$$

which, using $\mathbf{R}(\mathbf{u}^*) = 0$, leads nicely to:

$$\frac{\partial \mathbf{U}}{\partial t} = D\nabla^2\mathbf{U} + \mathbf{J}\mathbf{U}.$$

Here, we take the ansatz (justified in addendum):

$$\mathbf{U}(\mathbf{x},t) = \mathbf{v}e^{\lambda t + i\omega\cdot\mathbf{x}}, \qquad \lambda \in \mathbb{C},\ \mathbf{v} \in \mathbb{C}^2,\ \omega \in \mathbb{R}^3, \tag{5}$$

and we arrive at the following;

Proposition 3.3 *A homogeneous steady state $\mathbf{u}^* = (u^*, v^*)$ of the 2-component reaction-diffusion system (4), exhibits Turing instability under a perturbation of the form (5) if:*

1. $\text{Tr}\ \mathbf{J} = \partial_u R_U(\mathbf{u}^*) + \partial_v R_V(\mathbf{u}^*) < 0$, *and* $\det \mathbf{J} = \partial_u R_U(\mathbf{u}^*)\partial_v R_V(\mathbf{u}^*) - \partial_v R_U(\mathbf{u}^*)\partial_u R_V(\mathbf{u}^*) > 0$;

2. *There exists* $\omega \in \mathbb{R}^3$, *with* $\omega \neq 0$, *such that* $\mathbf{J} - |\omega|^2 D$ *has an eigenvalue* λ *with* $\Re(\lambda) > 0$.

Proof Substituting the ansatz into the linearised system yields:

$$\begin{aligned} \frac{\partial}{\partial t}(\mathbf{v}e^{\lambda t + i\omega\cdot\mathbf{x}}) &= D\nabla^2(\mathbf{v}e^{\lambda t+i\omega\cdot\mathbf{x}}) + \mathbf{J}(\mathbf{v}e^{\lambda t+i\omega\cdot\mathbf{x}}), \\ \lambda(\mathbf{v}e^{\lambda t+i\omega\cdot\mathbf{x}}) &= -D|\omega|^2(\mathbf{v}e^{\lambda t+i\omega\cdot\mathbf{x}}) + \mathbf{J}(\mathbf{v}e^{\lambda t+i\omega\cdot\mathbf{x}}), \\ \lambda\mathbf{v} &= (\mathbf{J} - D|\omega|^2)\mathbf{v}, \end{aligned}$$

which is now an eigenvalue problem.

Recalling Definition 3.1, we need the system to be temporally stable in the absence of diffusion, $D = 0$. So:

$$\lambda\mathbf{v} = \mathbf{J}\mathbf{v}.$$

Now this has the solution:

$$\lambda = \frac{\text{Tr}(\mathbf{J}) \pm \sqrt{(\text{Tr}(\mathbf{J})^2 - 4\det(\mathbf{J})}}{2}.$$

For stability we require $\mathbf{U}$ to decay in time, so we need $e^{\lambda t} \to 0$ as $t \to \infty$, which necessitates $\Re(\lambda) < 0$. As derived in [2, p. 302], this is given by:

$$\text{Tr}(\mathbf{J}) < 0, \qquad \det(\mathbf{J}) > 0.$$

Since $\Re(\lambda) = \frac{1}{2}\text{Tr}(\mathbf{J}) \pm \frac{1}{2}\Re\left(\sqrt{\text{Tr}(\mathbf{J})^2 - 4\det(\mathbf{J})}\right)$, and $\text{Tr}(\mathbf{J}) < 0$, $\det(\mathbf{J}) > 0$ is necessary for this.

Now, we want a $\omega \neq 0$, such that the perturbation grows in time with diffusion, as opposed to decaying. This is given by having $e^{\lambda t}e^{i\omega\cdot\mathbf{x}} \to \infty$ as $t \to \infty$. So, we need $\Re(\lambda) > 0$, where λ solves $\lambda\mathbf{v} = (\mathbf{J} - D|\omega|^2)\mathbf{v}$. □

We now have a way to quantitively determine instability criteria for reaction-diffusion systems.

Example (Gray Scott Analysis) Returning to our Gray Scott model:

$$\frac{\partial \mathbf{u}}{\partial t} = \begin{pmatrix} D_U & 0 \\ 0 & D_V \end{pmatrix} \nabla^2\mathbf{u} + \begin{pmatrix} -uv^2 + F(1-u) \\ uv^2 - (F+k)v \end{pmatrix}.$$

We then need the Jacobian of our rate function at the steady state (u^*, v^*), given by:

$$\mathbf{J} = \begin{pmatrix} -(v^*)^2 - F & -2u^*v^* \\ (v^*)^2 & 2u^*v^* - (F+k) \end{pmatrix},$$

and we now want the values of ω leading to eigenvalues with a positive real component of $\mathbf{J} - D|\omega|^2$.
So, we need eigenvalues of:

$$A = \begin{pmatrix} -(v^*)^2 - F - |\omega|^2 D_U & -2u^*v^* \\ (v^*)^2 & 2u^*v^* - (F+k) - |\omega|^2 D_V \end{pmatrix}.$$

This requires choosing some coefficient values — here (with choices becoming more apparent later), we choose $D_U = 0.16$, $D_V = 0.003$, $F = 0.04$, $k = 0.06$.

$$A = \begin{pmatrix} -(v^*)^2 - 0.04 - 0.16|\omega|^2 & -2u^*v^* \\ (v^*)^2 & 2u^*v^* - 0.1 - 0.003|\omega|^2 \end{pmatrix}.$$

We also need our steady state (u^*, v^*), which we can then calculate by solving the kinetic system at 0:

$$\begin{aligned} -u^*(v^*)^2 + 0.04(1 - u^*) &= 0, \\ u^*(v^*)^2 - 0.1v^* &= 0. \end{aligned}$$

A Python implementation to solve the eigenvalue problem yields steady state $u^* = 0.1158, v^* = 0.8633$, and Figure 3.

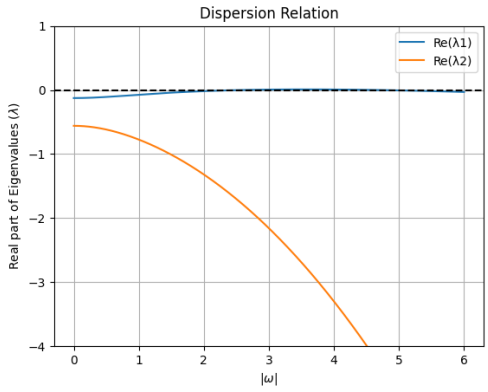


Figure 3: Plotting values $|\omega|$ against real components of eigenvalues $\Re(\lambda)$.

Looking at this figure, note that we see values around $2.5 < |\omega| < 4$ giving positive values of $\Re(\lambda)$. This should tell us that we have a system that exhibits Turing instability, and after a simulation, we get Figure 4.

Which gives us exactly what we'd expect — a self organised pattern — in this case, a stripy-labyrinthine pattern — which we'll later see is consistent with our band of unstable modes.

Note also, that despite the fact that in this case, we picked our parameters seemingly without much reason, if we're given 2 of the 4 initial

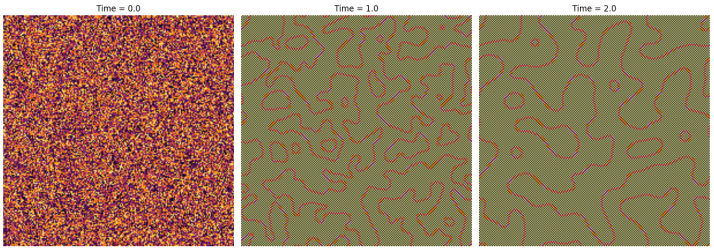


Figure 4: A Gray Scott simulation using the coefficient values chosen earlier.

parameters, we can make a phase portrait to work out which values of the other 2 will give us a Turing instability by checking eigenvalues.

As an example, suppose we're given the diffusive coefficients D_U, D_V, then we can create a plot as in Figure 5.

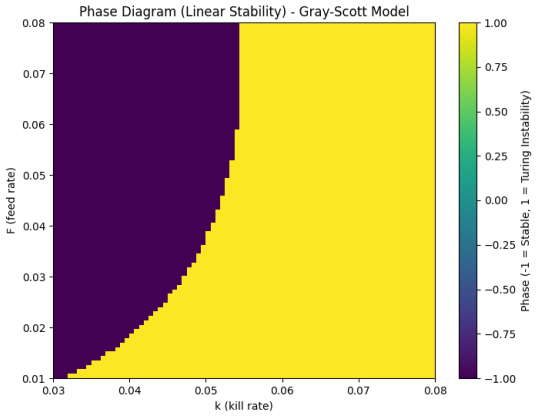


Figure 5: A Gray Scott phase-diagram using the diffusive coefficients $D_U = 0.16, D_V = 0.003$.

4 Pattern Classification

It now becomes natural to think about how we can begin to classify these patterns. Of course, we know that in nature, spots, stripes, intricate maze-like designs and regular shapes are all somewhat commonplace.

So here, we restrict ourselves to systems in 2 spatial dimensions — with hopes of finding criterion for forming particular patterns.

4.1 Classification via Dispersion Relation

Returning to our dispersion relation, we've motivated the following definition:

Definition 4.1 *Let $\lambda(\omega) \in \mathbb{C}$ be the eigenvalue(s) corresponding to a particular mode $\omega \in \mathbb{R}^3$ as in part 2 of Proposition 3.3. Then, we define:*

$$\lambda_\omega = \{\omega \in \mathbb{R}^3 : \Re(\lambda(\omega)) > 0\},$$

as the set of all unstable modes.

In [5, p. 5] Krause et al. note that "it is often the case that the band of unstable modes is large" and "in such cases, these modes will compete with one another, and one cannot say apriori what wavelength the final pattern might have".

This then motivates 2 main cases for the unstable modes in λ_ω, one with a large band of unstable modes, and one with a single dominating mode:

1. We have a single mode $\omega^* \in \lambda_\omega$ with $\Re(\lambda(\omega^*))$ much bigger than the others — i.e. a large dominating unstable mode;
2. We have multiple modes $\omega_i \in \lambda_\omega,\ i = \{1, 2, \ldots, n\},\ n \in \mathbb{N}$, with similarly sized $\Re(\lambda(\omega))$ — i.e. a band of unstable modes.

4.2 Single Large Unstable Mode

For a single large unstable mode ω^*, we can then consider 2 subcases; where $\Im(\lambda(\omega^*)) = 0$, and where $\Im(\lambda(\omega^*)) \neq 0$.

Lemma 4.2 *Suppose there exists a unique mode $\omega^* \in \lambda_\omega$ with $\Im(\lambda(\omega^*)) = 0$ such that:*

$$\Re(\lambda(\omega^*)) > \Re(\lambda(\omega)), \quad \forall \omega \in \lambda_\omega,\ \omega \neq \omega^*.$$

Then for sufficiently large time, the solution is dominated by a single mode:

$$U(\mathbf{x}, t) \sim Ae^{\lambda(\omega^*)t}e^{i\omega^* \cdot \mathbf{x}}.$$

for some $A \in \mathbb{C}$

Proof From Addendum 6.1, we know we can represent any perturbation in the form $\mathbf{U}(\mathbf{x}, t) = \int \hat{U}(\omega, t)e^{i\omega \cdot \mathbf{x}}\ \mathrm{d}\omega$.

We also from the Addendum get $\partial_t \hat{U}(\omega, t) = (\mathbf{J} - |\omega|^2 D)\hat{U}(\omega, t)$, which is a linear ODE, solving neatly to $\hat{U}(\omega, t) = e^{\lambda(\omega)t}U(\omega, 0)$.

Now, if $\Re(\lambda(\omega^*)) > \Re(\lambda(\omega))$ for all $\omega \neq \omega^*$, then as $t \to \infty$;

$$\mathbf{U}(\mathbf{x}, t) \approx \hat{U}(\omega^*, 0)e^{\lambda(\omega^*)t}e^{i\omega^* \cdot \mathbf{x}}.$$

Exactly as required. □

A single dominating mode means that our solution will have repetitive structures throughout the domain. This exactly describes the formation of stripes.

Example (Gray Scott - Single Large Real Mode) We pick the initial parameters $D_U = 0.16, D_V = 0.003, F = 0.02, k = 0.1$ and get the dispersion relation in Figure 6, and the simulation as expected in Figure 7.

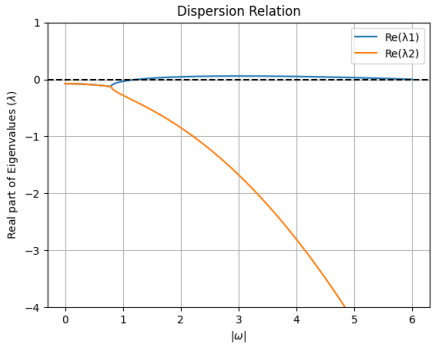


Figure 6: Dispersion relation of the example detailed above.

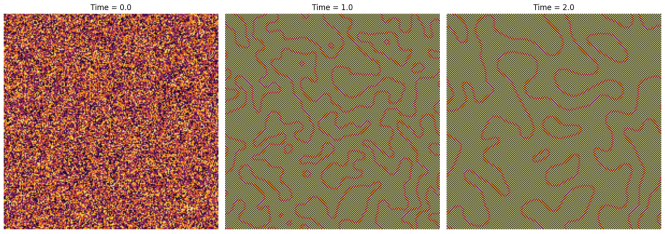


Figure 7: Simulation of the example detailed above.

Lemma 4.3 *Suppose there exists $\omega^* \in \mathbb{R}^3$ such that for $\alpha \in \mathbb{R}$ and $\beta \in \mathbb{R}$:*

$$\lambda(\omega^*) = \alpha + i\beta, \quad \alpha > 0, \ \beta \neq 0.$$

Then the corresponding mode evolves as

$$U(x,t) \sim Ae^{\alpha t}e^{i(\omega^* \cdot \mathbf{x} + \beta t)},$$

for some $A \in \mathbb{C}$.

Proof This proof is exactly the same as before. But in this case, where $\mathbf{U}(\mathbf{x},t) \approx \hat{U}(\omega^*, 0)e^{\lambda(\omega^*)t}e^{i\omega^* \cdot \mathbf{x}}$ as $t \to \infty$, we note that $\lambda(\omega^*) = \alpha + i\beta$, so:

$$\mathbf{U}(\mathbf{x},t) \approx \hat{U}(\omega^*, 0)e^{(\alpha+i\beta)t}e^{i\omega^* \cdot \mathbf{x}} = Ae^{\alpha t}e^{i(\omega^* \cdot \mathbf{x} + \beta t)},$$

exactly as required. □

Now, this case is fundamentally different to that of Lemma 4.2. We also have here a temporal oscillation with $e^{\beta t}$. Here, [1, pp. 261-262] helps us explain what is happening here.

When both a Turing instability (real positive eigenvalue at some $\omega^* \neq 0$) and a Hopf instability (complex eigenvalue at $\omega = 0$) — a type of temporal instability — occur simultaneously or near-simultaneously, the two interact. This is called a Turing-Hopf bifurcation. Near such a codimension-2 bifurcation point (where both instability conditions are met), the solution can be written as:

$$U(x,t) = A(t)e^{i\omega^* \cdot \mathbf{x}} + B(t),$$

where $B(t)$ is the amplitude of the Hopf mode, and $A(t)$ the amplitude of the Turing mode.

This presents itself as travelling waves, stationary patterns, or chaotic regimes depending on the relative strengths of A and B.

4.3 Large Band of Unstable Modes

Now, in the context of large bands of similarly unstable modes, we noted earlier that it is not necessarily possible to determine the characteristics of the final pattern.

However, we could simplify this case by considering a band of dominating unstable modes that present some sort of symmetry. For example, as described in [8, p. 12766] we could consider 3 modes $\omega_i,\ i = \{1,2,3\}$ with $\sum_{i=1}^{3} \omega_i = 0$, $|\omega_1| = |\omega_2| = |\omega_3| = \omega^*$ and $\lambda(\omega_i) = \lambda^*$, as these modes interact with each other in very predictable way, given that they grow at the same rate and travel in useful, symmetrical directions.

For this case, we'll work in 2 spatial dimensions for simplicity.

Lemma 4.4 *Suppose there exist distinct $\omega_1, \omega_2, \omega_3 \in \mathbb{R}^2$ such that*

$$|\omega_1| = |\omega_2| = |\omega_3| = \omega^*, \quad \omega_1 + \omega_2 + \omega_3 = 0,$$

and $\mathrm{Re}(\lambda(\omega_i)) = \lambda^* > 0$ for $i = 1,2,3$, with $\mathrm{Re}(\lambda(\omega)) < \lambda^*$ for all other ω. Then the solution admits a leading-order approximation of the form

$$U(\mathbf{x},t) \sim \sum_{i=1}^{3} A_i(t)e^{i\omega_i \cdot \mathbf{x}},$$

for some $A_i(t) \in \mathbb{C}$. Moreover, for real and equal amplitudes, this approximation is periodic with respect to a hexagonal lattice.

Proof The leading-order approximation follows by the same argument as Lemma 4.2, with three dominating modes in place of one. It remains to show that the resulting spatial pattern is hexagonal.

We take $\mathbf{x} \in \mathbb{R}^2$ and choose the three modes explicitly. The conditions $|\omega_i| = \omega^*$ and $\omega_1 + \omega_2 + \omega_3 = 0$ are satisfied by:

$$\omega_1 = \omega^* \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \omega_2 = \omega^* \begin{pmatrix} -\frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{pmatrix}, \quad \omega_3 = \omega^* \begin{pmatrix} -\frac{1}{2} \\ -\frac{\sqrt{3}}{2} \end{pmatrix},$$

which are unit vectors (up to scaling by ω^*) separated by angles of $120°$, and with $\omega_1 + \omega_2 + \omega_3 = \mathbf{0}$.

Taking all amplitudes A_i to be real and equal, the real part of the approximation gives the spatial pattern

$$U(\mathbf{x}) = \sum_{i=1}^{3} \cos(\omega_i \cdot \mathbf{x}).$$

We claim that U is periodic with respect to the lattice Λ generated by the vectors

$$\mathbf{a}_1 = \frac{2\pi}{\omega^*} \begin{pmatrix} 1 \\ \frac{1}{\sqrt{3}} \end{pmatrix}, \quad \mathbf{a}_2 = \frac{2\pi}{\omega^*} \begin{pmatrix} 0 \\ \frac{2}{\sqrt{3}} \end{pmatrix}.$$

To verify this, it suffices to check that $U(\mathbf{x} + \mathbf{a}_j) = U(\mathbf{x})$ for $j = 1, 2$, which holds if and only if $\omega_i \cdot \mathbf{a}_j \in 2\pi\mathbb{Z}$ for all i, j by periodicity of cos. Computing:

$$\begin{aligned}
\omega_1 \cdot \mathbf{a}_1 &= \omega^* \cdot \frac{2\pi}{\omega^*} \cdot = 2\pi, \\
\omega_2 \cdot \mathbf{a}_1 &= \omega^* \left(-\frac{1}{2} \cdot \frac{2\pi}{\omega^*} + \frac{\sqrt{3}}{2} \cdot \frac{2\pi}{\omega^* \sqrt{3}} \right) = \omega^* \cdot \frac{2\pi}{\omega^*} \left(-\frac{1}{2} + \frac{1}{2} \right) = 0, \\
\omega_3 \cdot \mathbf{a}_1 &= \omega^* \left(-\frac{1}{2} \cdot \frac{2\pi}{\omega^*} - \frac{\sqrt{3}}{2} \cdot \frac{2\pi}{\omega^* \sqrt{3}} \right) = \omega^* \cdot \frac{2\pi}{\omega^*} \left(-\frac{1}{2} - \frac{1}{2} \right) = -2\pi,
\end{aligned}$$

and similarly $\omega_1 \cdot \mathbf{a}_2 = 0$, $\omega_2 \cdot \mathbf{a}_2 = 2\pi$, $\omega_3 \cdot \mathbf{a}_2 = -2\pi$, all of which lie in $2\pi\mathbb{Z}$, confirming periodicity with respect to Λ.

The lattice Λ generated by $\mathbf{a}_1$ and $\mathbf{a}_2$ is a standard hexagonal lattice. □

Note the contrast with the single-mode case of Lemma 4.2. A single mode $\cos(\omega^* \cdot \mathbf{x})$ is periodic in one spatial direction only, producing stripes. The three-mode combination breaks this to periodicity in two independent directions at $60°$ to each other, which is the defining geometric property of a hexagonal lattice, and hence produces hexagonal patterns.

5 Conclusion

We've now successfully built tools for understanding how and when these patterns form in reaction-diffusion systems, and how we can understand better the conditions required to produce particular patterns.

But, some lingering ideas remain. We may not want spontaneous patterns to form. For example, in a battery, non-uniform ion concentration near electrodes has the potential to amplify perturbations in the otherwise uniform, homogeneous chemical mixture.
So, we ask the question; what can we do to minimise the effects of diffusion driven instability — and more precisely — can we add another reactant, or a particular physical structure to limit pattern formation.

There are many such problems and questions in the study of Turing instabilities and reaction-diffusion systems, and equally many applications that make the study of such systems crucial.

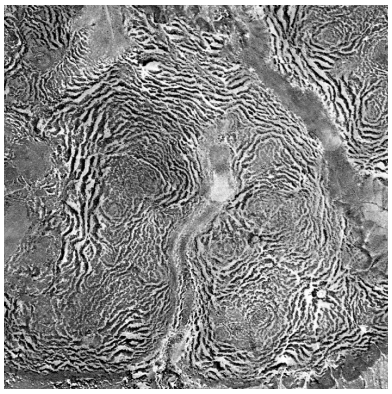


Figure 8: The "tiger bush" phenomenon - taken from the USGS

The tiger bush is a famous example of emergent Turing patterns. By understanding exactly how the tiger bush pattern forms, we can control desertification, manage water, and restore fragile ecosystems.

6 Addendum

6.1 Justification of Ansatz (5)

Now we need to think about what form we want our perturbation to take. Note however, that whatever our perturbation is, it's essentially changing concentrations of chemicals at certain spatial points, and then allowing these changes in concentration to diffuse throughout the medium — interacting with existing populations of chemical, and each other — in essence, we have something like a ripple effect; a single change spreading throughout the medium.

In general, any small perturbation could be decomposed using Fourier transforms into:

$$\mathbf{U}(\mathbf{x},t) = \int \hat{U}(\omega,t)e^{i\omega\cdot\mathbf{x}}\,\mathrm{d}\omega, \qquad \omega \in \mathbb{R}^3.$$

With $\hat{U}$ the Fourier coefficients. But of course, in a linearised system, these different Fourier modes don't interact with each other, since:

$$\begin{aligned} \frac{\partial \mathbf{U}}{\partial t} &= \int \partial_t \hat{U}(\omega,t)e^{i\omega\cdot\mathbf{x}}\,\mathrm{d}\omega, \\ \nabla^2\mathbf{U} &= \int -\hat{U}(\omega,t)|\omega|^2 e^{i\omega\cdot\mathbf{x}}\,\mathrm{d}\omega, \\ \text{so,} & \\ \int \partial_t \hat{U}(\omega,t)e^{i\omega\cdot\mathbf{x}}\,\mathrm{d}\omega &= \int (\mathbf{J} - |\omega|^2 D)\hat{U}(\omega,t)e^{i\omega\cdot\mathbf{x}}\,\mathrm{d}\omega. \end{aligned}$$

So for each ω:

$$\partial_t \hat{U}(\omega,t) = (\mathbf{J} - |\omega|^2 D)\hat{U}(\omega,t).$$

So, since different modes don't interact with each other (each $e^{i\omega\cdot\mathbf{x}}$ only maps to itself) — it's enough for us to consider just a single mode:

$$\mathbf{U}(\mathbf{x},t) = \hat{U}(\omega,t)e^{i\omega\cdot\mathbf{x}}.$$

Note further that:

$$\partial_t \hat{U}(\omega,t) = (\mathbf{J} - |\omega|^2 D)\hat{U}(\omega,t),$$

is just a linear ODE with constant coefficients (no $\mathbf{x}$ terms).

And so, as detailed in [2, p. 35], we can take the ansatz $\hat{U}(\omega,t) = \mathbf{c}_\omega e^{\lambda t}$ where $\lambda \in \mathbb{C}$ and $\mathbf{c}_\omega \in \mathbb{C}^2$.
From which we arrive at:

$$\mathbf{U(x,t)} = \mathbf{c}_k e^{\lambda t + i\omega\cdot\mathbf{x}}.$$

This ansatz can be thought of as probing the system with a single wave with wavelength determined by ω, and changes in time determined by λ allowing us to determine how perturbations scale change in time.

6.2 Code

All Python code written for simulations, diagrams and phase portraits can be found in the following google drive folder:
https://drive.google.com/drive/folders/1kmvCAh0hBhjJGdWAXCvzU27Gxt859bWS

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